From Farm and Weather Data to Solar and Battery Trading Decisions

Data, forecast paths and auction participation

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Abstract

A solar producer must commit electricity before its production is known. Battery storage provides flexibility, while successive auctions create opportunities to revise the contracted position as weather information improves. We study this problem using four northern-Italian farms, historical production and irradiance, regularly updated weather forecasts, and 50-member weather ensembles. Weather is converted into possible AC production paths; these paths support physically feasible trading and battery decisions. Participation studies compare every nonempty subset of the day-ahead and three intraday auctions, with each strategy carrying its own state through time. This document explains the available data, the forecast-to-decision process, and the first participation results. The full optimization and supporting equations are collected in appendices.

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1 The problem and our core idea

A photovoltaic farm and a battery share a grid connection. Solar generation can be exported, stored or curtailed; the battery can also charge from grid imports. The producer sells and buys energy in the Italian day-ahead auction (DA) and three intraday auctions (IDA1, IDA2 and IDA3). An order becomes binding when submitted. Later auctions can adjust the resulting position, while battery operation helps deliver it under uncertain solar production.

Our core idea is to connect the information available at each auction with *the decisions that can still be changed*. We transform eligible weather forecasts into production paths, optimize the remaining market opportunities and battery operation together, submit the current order, and repeat as new information arrives. Earlier commitments remain fixed. The actual battery state links the decisions across delivery intervals and days.

The model maximizes expected remaining auction revenue less battery wear and trading costs, subject to energy conservation, exclusive charging/discharging, stored-energy bounds and the shared grid limit. The full optimization is in Appendix A.

This allows us to ask three concrete questions. How does participation in additional auctions change the achievable operating outcome? What changes when a single production forecast is replaced by an ensemble? How much does joint planning of remaining auctions contribute relative to optimizing one auction at a time? We assess profit together with imbalance, curtailment, wear and ending stored energy.

2 The available data

2.1 Four farms and their historical records

The dataset covers four anonymous farms in the northern-Italian bidding zone. Installed power is measured on the AC side. The supplied metadata describe fixed arrays and single-axis trackers:

Farm	AC capacity (MW)	Declared technology	Battery (MWh)	Power (MW)
1	0.95328	Fixed array	0.75	0.5
345	1.52460	Fixed array	1.00	0.5
901	4.16000	Bifacial single-axis tracker	2.00	1.0
910	6.63800	Single-axis tracker	5.00	2.5

The batteries are hypothetical study assets. Their energy and power ratings, 7–93% state-of-charge range, 90% round-trip efficiency and connection limits are modeling assumptions. The farm metadata also provide nominal orientation, tilt and elevation. Exact farm coordinates are unavailable, so solar-position calculations use documented geographical proxies; fitted effective geometry or rear-irradiance assumptions can differ from the metadata label.

Historical records run from 1 January 2025 to 31 August 2026 on a 15-minute UTC grid: 58,368 timestamps for each farm. They contain AC power, historical satellite global horizontal irradiance (GHI), and historical diffuse fraction. This is the time grid's size, rather than a count of independently verified, complete measurements. Power is converted once from kW to MW; multiplying by the interval duration gives energy in MWh.

Production quality and availability matter to both forecasting and trading. Reviewed modeling views distinguish retained measurements, approved synthetic replacements, seasonal night zeros and excluded intervals. Historical-GHI reconstructions provide explicitly identified counterfactual production where used. Raw historical data, reviewed targets and reconstructed production have

different evidential meanings. Forecast training and replay therefore retain their target policy and provenance.

2.2 Updated deterministic weather forecasts

The main weather archive contains 2,440 initialization times, every six hours, from 30 December 2024 to 31 August 2026. Each initialization supplies 49 hourly lead coordinates, from zero to 48 hours, for every farm. The weather fields include global and direct horizontal radiation, total and layered cloud cover, air temperature, dew point, wind components at 10 and 100 metres, precipitation, snow-water equivalent and visibility.

A forecast is identified by its initialization and lead time; their sum gives the valid time. Several issues can predict the same delivery interval. This provides a record of forecast revision as delivery approaches. The four daily issues are successive information updates, each supplying one deterministic weather trajectory.

Hourly radiation represents an interval mean, and precipitation an hourly accumulation. Alignment with the quarter-hour delivery grid respects those semantics. Issue time, assumed publication time and delivery time are retained separately. In the studies, forecasts become available six hours after initialization; this is an explicit release assumption.

2.3 Weather ensembles

Two separate archives supply *50 weather members for the same issue and lead times*, with the same four farms and 0–48-hour lead coordinates:

Archive	Available 00 UTC initializations	Members / hourly leads
Winter	31 Dec. 2025–31 Jan. 2026 (32 issues)	50 / 49
Late summer	1 Aug.–30 Sept. 2026 (61 issues)	50 / 49

These members represent alternative weather trajectories within one forecast issue. Keeping a member's identity through the horizon retains its supplied temporal evolution, including the duration of cloudy and sunny periods. The spread supplies uncertainty information that a single deterministic forecast cannot describe. Equal scenario weights are a modeling choice; coverage and calibration require separate assessment.

The participation studies freeze the ensemble initialized at 00 UTC on the *preceding delivery day*. The winter archive includes 31 December, which supports 1 January delivery. The late-summer archive begins on 1 August and therefore supports this convention from 2 August. Its September weather issues extend beyond the available farm-production record used here.

2.4 Market observations

Historical clearing curves cover DA and the three intraday auctions, with imbalance-settlement prices used to value actual delivery deviations. Native product durations and daylight-saving changes are preserved. Physical operation uses quarter-hours; earlier hourly DA products remain single traded products when mapped onto that finer grid.

The participation comparisons use actual future auction prices as a common price-information benchmark. This isolates choices about market access, production forecasts and physical flexibility under that benchmark. Their profits must be interpreted with this price-foresight assumption. Developing price forecasts is a further route toward operational decisions.

3 From weather information to production and decision paths

3.1 Selecting information at the auction cutoff

The study uses the following market timeline, in Italian local time. Decisions are taken five minutes before each gate:

Auction	Gate	Delivery covered	Latest typical weather issue
DA	Previous day, 12:00	Whole next day	Previous day, 00 UTC
IDA1	Previous day, 15:00	Whole next day	Previous day, 06 UTC
IDA2	Previous day, 22:00	Whole next day	Previous day, 12 UTC
IDA3	Delivery day, 10:00	Noon to midnight	Delivery day, 00 UTC

The last column follows the six-hour release assumption and requires sufficient remaining-horizon coverage. At every cutoff, the deterministic forecast is updated to the latest eligible issue. The 50 ensemble paths retain the frozen preceding-day issue, so their information age differs from the updating forecast. This distinction is part of the comparison design.

3.2 Converting weather into AC generation

The physical forecast first derives direct and diffuse irradiance components, then projects them onto the array or tracker plane. It estimates module temperature from air temperature, wind and plane-of-array irradiance, computes DC production and converts it to AC power. Output is bounded by the farm AC rating and forced to zero during the specified seasonal nighttime windows. Effective conversion parameters can be estimated from historical data.

Statistical and neural models provide another mapping from weather, calendar, lead-time and farm information to AC production. Historical power and irradiance support fitting and diagnosis; evaluation uses the forecast weather available for the delivery interval. Where learning uses historical irradiance, the shift to forecast irradiance is stated explicitly. Chronological training, selection and evaluation periods are kept distinct. Measured-only and reconstructed-target scores answer different questions.

Each ensemble weather member passes separately through the selected conversion model, producing a complete AC power path on the delivery grid. The study's 51-path input combines these 50 paths with the updating deterministic forecast. This ensemble construction differs from generating paths by independently sampling marginal quantiles at each time. The full path matters to a battery: a sustained production shortfall can exhaust stored energy even when its pointwise forecast spread resembles that of a shorter shortfall.

3.3 Turning forecast paths into a trading policy

At a given cutoff, a production path is a possible external input. A decision path consists of auction quantities and feasible battery/solar actions along the represented information history. Scenarios with the same information must share the decisions made before they can be distinguished. Future production is therefore not disclosed merely by representing several paths.

In the strict ensemble participation studies, orders and planned battery commands are common across all 51 production paths; curtailment can adapt to available production. Every represented path must support the contracted delivery. With identical price paths and the stated curtailment recourse, useful solar injection is bounded by the pointwise minimum across production paths. This makes the strict formulation conservative. An ensemble of forecasts does not automatically imply adaptive future battery decisions.

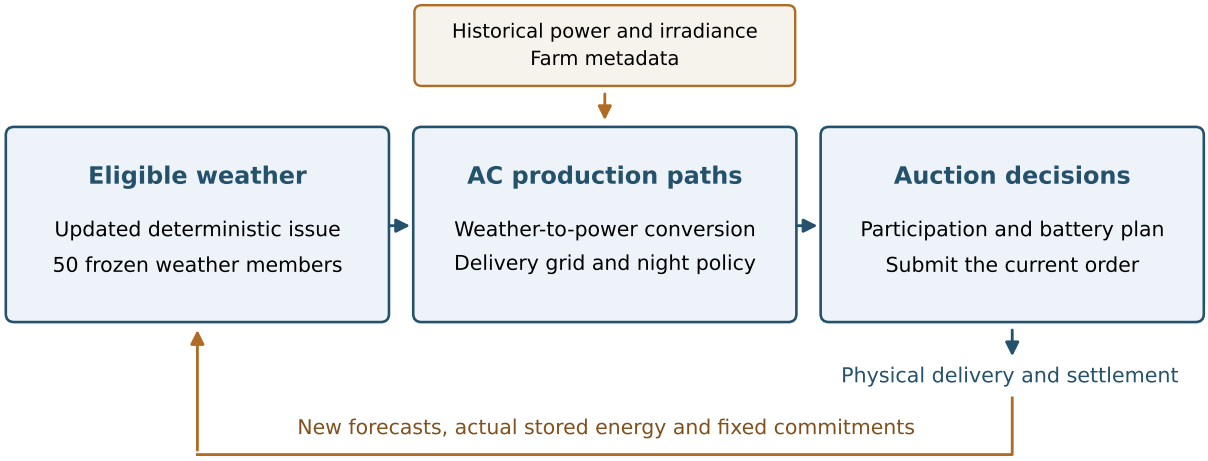


Figure 1: Historical records support the production mapping and evaluation. Eligible weather issues supply forecast paths; those paths support auction and physical decisions. Realized delivery supplies the carried battery state.

Integrated planning optimizes the remaining participating auctions together. Only the closing auction's quantity is submitted. At the next gate, that order is fixed and provisional later orders are reoptimized with new eligible forecasts. During delivery, commitment tracking adjusts physical battery power within its energy, power and grid limits. Actual SOC, submitted commitments, wear and transaction-fee state carry into subsequent decisions and days. A forecast projection supplies the starting state for tomorrow's plan while today's delivery is still in progress; the realized state governs execution.

4 The auction participation studies

4.1 What is compared

Four optional auctions give $2^4 - 1 = 15$ nonempty participation profiles: four single-auction profiles, six pairs, four triples and the all-auction profile. An auction omitted from a profile supplies no submission opportunity. The remaining gates retain their chronological order and delivery eligibility.

Every profile has its own continuous physical and financial trajectory. At the start of the study the initial state is common within a farm; afterward each strategy's actions determine its own SOC and later feasible trades. The January 2026 comparison starts at 90% SOC and carries state over 31 days. Planning uses a 50% terminal target and a 40% floor; forecast error and physical execution can yield different realized ending inventories.

The study runs each profile with an updating deterministic production forecast and with the 51-path input. Two references clarify the mechanism:

- **Sequential all-auction planning:** participate in every auction, but optimize only the current auction's quantities at each gate. Later gates can still revise the physical plan and add their own trades.
- **Perfect-production reference:** disclose future production while retaining the same asset and actual-price benchmark. It shows the outcome of that information counterfactual under its specified planning convention.

Integrated and sequential planning thus differ even when they have the same auction access. With price foresight, the integrated model can anticipate cross-auction spreads and coordinate earlier purchases/sales with later ones. The resulting difference combines that economic opportunity with forecast and physical effects.

4.2 First results and their interpretation

An earlier January 2025 farm-910 study compared all 15 profiles on independently initialized days. Its all-auction integrated profile ranked first, with mean profit EUR 3,520.06/day and total EUR 109,121.72 over 31 days. The continuous January 2026 studies then retained the actual trajectory across days for all four farms and both forecast variants. These two designs have different state assumptions and are evaluated separately.

The following January 2026 cash totals summarize three key participation choices. D denotes deterministic production and E the strict 51-path input. All columns use the same actual-price benchmark:

Farm	DA only D EUR	Sequential all D EUR	Integrated all D EUR	Integrated all E EUR
1	1,648.99	1,848.22	10,216.33	10,197.54
345	6,494.09	6,916.97	20,306.16	18,555.04
901	16,467.77	16,904.75	53,708.73	49,397.35
910	19,804.62	20,500.95	78,971.91	76,171.83

The large integrated/sequential difference is consistent with the ability to plan actual-price spreads across auctions. It cannot be attributed solely to more accurate solar forecasts. Figure 2 shows the full profile comparison for farm 1; it also makes the contribution of particular auction combinations visible.

Strict ensemble planning reduces absolute imbalance for the all-auction profile on all four farms. For farm 1 it changes 5.818 to 1.577 MWh, with wear EUR 520.52 versus EUR 525.84 and final SOC 33.50% versus 43.49%. For farms 345, 901 and 910, absolute imbalance changes from 2.447 to 0.780, 23.099 to 4.320, and 70.412 to 21.643 MWh respectively. Cash profit decreases in each all-auction comparison. This illustrates how uncertainty representation changes market exposure and use of physical flexibility; reducing imbalance alone does not determine the economically preferred profile.

Separate full January and August production-input comparisons provide winter and summer cases; August starts on the second because the required preceding ensemble issue is absent for its first day. Each month starts independently. Reported outcomes include daily profit distributions, settlement cash, curtailment, charge/discharge, wear and ending SOC. Cash profit excludes capital cost and terminal inventory valuation. The rankings are retrospective and use actual-price foresight; historical model selection and the target provenance further limit an operational interpretation.

5 What this framework enables next

The available data connect forecast revision, production uncertainty and sequential participation within one physical chronology. The next questions are how forecast calibration and price prediction change those participation results, and how receding-horizon battery control uses new observations between auction gates. These developments build on the same data-to-path-to-decision structure. Longer comparisons must retain common dates, target provenance, information timing and ending-inventory accounting.

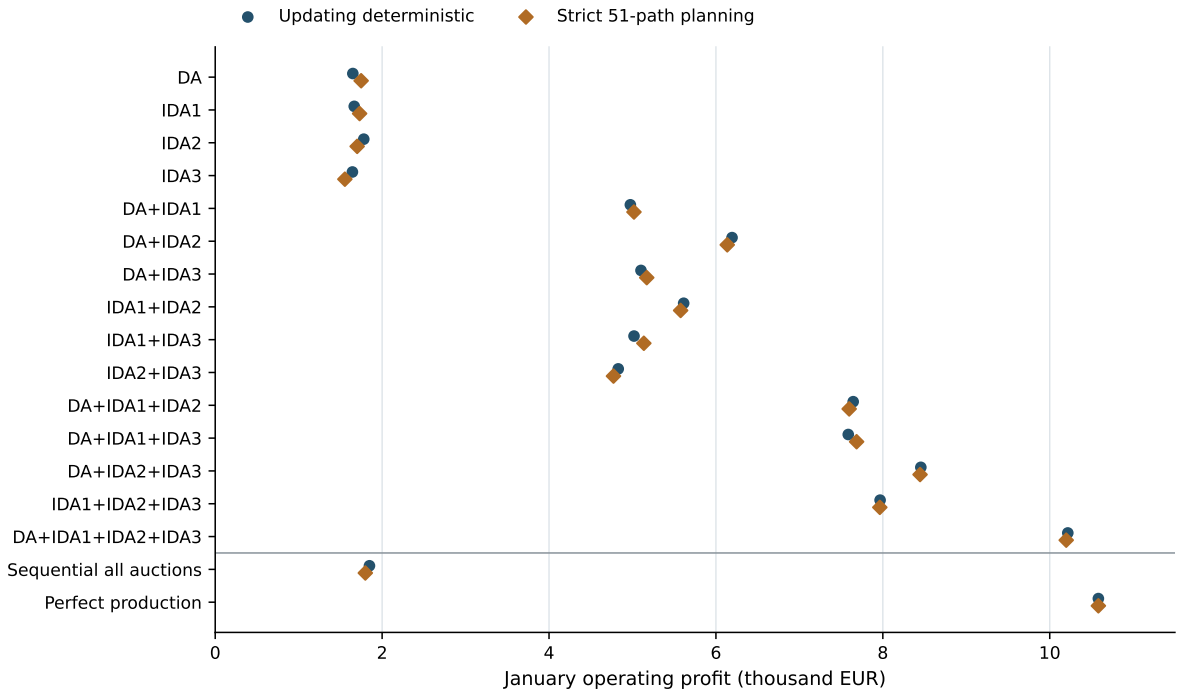


Figure 2: Farm 1, continuous January 2026: monthly operating profit for all 15 auction subsets and the two references. Dots show the deterministic and strict 51-path inputs. Earlier orders remain fixed and each profile carries its own battery state. Perfect production is a separately labelled hindsight reference.

A Full residual auction optimization

A.1 Sets, parameters and signs

Let $t \in \mathcal{T}$ index the remaining physical intervals of a delivery day, Δ_t their duration in hours, $s \in \mathcal{S}$ scenarios with probabilities w_s, and $a \in \mathcal{A}_k$ auctions still open at decision k. $\mathcal{T}_a$ is auction a's eligible delivery set. Native products group physical intervals; their identifiers retain daylight-saving offsets.

Solar availability is $V_{s,t}$ MW AC, battery capacity E MWh, power limit $\bar{P}$ MW, energy bounds $E^{\min}, E^{\max}$, efficiencies η_c, η_d and hourly self-discharge σ. The grid limits are $P^{\text{imp}}, P^{\text{exp}}$. Q_t^{fix} is the signed MWh position from all earlier submitted quantities, including still-unpriced orders. Positive trade/grid signs mean sale/export. Auction price coefficients are $\pi_{a,s,t}$ in EUR/MWh.

Controls $c_{n,t}, d_{n,t}$ are charging/discharging MW; $z_{n,t}$ is a charging binary. Curtailment is $u_{m,t}$ MW, injection $i_{s,t}$ MW, grid exchange $g_{s,t}$ MW and internal energy $e_{s,t}$ MWh. $q_{a,n,t}$ is a signed auction adjustment in MWh. Node maps $n_t(s), n_a(s), m_t(s)$ encode observable history. Leaf values below refer to their node-owned variables, so indistinguishable histories cannot take different actions.

A.2 Complete base problem

For throughput wear $D_s = c_{\text{thr}} \sum_t (c_{s,t} + d_{s,t})\Delta_t$ and optional import cost $I_s = f \sum_t [-g_{s,t}]_+ \Delta_t$, the residual model is

$$\max \quad \sum_s w_s \left[\sum_{a \in \mathcal{A}_k} \sum_{t \in \mathcal{T}_a} \pi_{a,s,t} q_{a,n_a(s),t} - D_s - I_s - C_s^{\text{trade}} \right] \tag{1}$$

$$\text{s.t.} \quad i_{s,t} + u_{m_t(s),t} = V_{s,t}, \quad i_{s,t} \geq 0, \quad u_{m_t(s),t} \geq 0, \qquad \forall s,t, \tag{2}$$

$$g_{s,t} = i_{s,t} + d_{n_t(s),t} - c_{n_t(s),t}, \qquad \forall s,t, \tag{3}$$

$$-P^{\text{imp}} \leq g_{s,t} \leq P^{\text{exp}}, \qquad \forall s,t, \tag{4}$$

$$0 \leq c_{n,t} \leq \bar{P} z_{n,t}, \quad 0 \leq d_{n,t} \leq \bar{P}(1 - z_{n,t}), \quad z_{n,t} \in \{0,1\}, \qquad \forall n,t, \tag{5}$$

$$e_{s,t+1} = (1-\sigma)^{\Delta_t} e_{s,t} + \eta_c c_{n_t(s),t} \Delta_t - d_{n_t(s),t} \Delta_t / \eta_d, \qquad \forall s,t, \tag{6}$$

$$E^{\min} \leq e_{s,t} \leq E^{\max}, \quad e_{s,0} = e^k, \qquad \forall s,t, \tag{7}$$

$$Q_{s,t} = Q_t^{\text{fix}} + \sum_{a : t \in \mathcal{T}_a} q_{a,n_a(s),t}, \qquad \forall s,t, \tag{8}$$

$$g_{s,t} \Delta_t = Q_{s,t}, \qquad \forall s,t, \tag{9}$$

$$-P^{\text{imp}} \Delta_t \leq q_{a,n,t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,n,t \in \mathcal{T}_a, \tag{10}$$

$$-P^{\text{imp}} \Delta_t \leq Q_t^{\text{fix}} + \sum_{b \leq a : t \in \mathcal{T}_b} q_{b,n_b(s),t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,s,t, \tag{11}$$

$$q_{a,n,t}/\Delta_t = q_{a,n,t'}/\Delta_{t'}, \qquad t,t' \text{ in the same native product.} \tag{12}$$

Only eligible quantities exist. The last equality ties average power within a native product; on equal quarter-hours it ties MWh shares directly. Fixed initial energy can be observed or a forecast projection to the eligible boundary. A configured terminal equality $e_{s,T} = e_F$ and/or floor $e_{s,T} \geq e_{\text{floor}}$ is added where the study requires it. A planning terminal target does not itself force an algebraic feedback controller to end at that realized energy.

For progressive transaction tariff F_m, previously submitted monthly gross volume V_m and new gross volume $G_{m,s} = \sum_{a : m(a) = m, t} |q_{a,n_a(s),t}|$,

$$C_s^{\text{trade}} = \sum_m \{F_m(V_m + G_{m,s}) - F_m(V_m)\}. \tag{13}$$

Exact absolute-value and piecewise-linear graphs are required if reachable marginal tariff bands are decreasing; a convex epigraph alone is insufficient. The month is the submission month, not

necessarily the delivery month. Earlier auction cash and incurred wear are sunk and absent from (1). Only the closing auction's order is submitted; other quantities are revisable plans. Sequential planning replaces $\mathcal{A}_k$ by the single current auction.

A.3 Exact envelope for the strict ensemble problem

Under exact delivery, common trades/battery controls, identical price paths and compatible instantaneous curtailment recourse, i_t is shared and

$$0 \le i_t \le \min_s V_{s,t}, \qquad u_{s,t} = V_{s,t} - i_t. \tag{14}$$

This gives an exact reduced feasible set and objective. It does not apply to general adaptive or imbalance-enabled problems, and discarding low-probability paths is not part of the equivalence.

B Battery wear, control and realized accounting

B.1 Wear models

For replacement value R, reference internal cycle depth d_0, life N and capacity E, AC throughput wear can be calibrated as

$$c_{\text{thr}} = \frac{R}{N d_0 E(1/\eta_c + \eta_d)}, \qquad D = c_{\text{thr}} \sum_t (c_t + d_t)\Delta_t. \tag{15}$$

An optional segmented depth model uses stress $\Phi(d) = \alpha d^\beta$, $\beta \ge 1$, J segments and marginal discharge cost

$$m_j = \frac{RJ}{\eta_d E}\{\Phi(j/J) - \Phi((j-1)/J)\}, \qquad D = \sum_{t,j} m_j d_{j,t}\Delta_t. \tag{16}$$

Segment energy transitions sum to aggregate energy; shallow-first charging and discharging priorities preserve cycle-depth memory. The segment vector carries between decisions and cannot generally be recovered from SOC alone. This is an approximation to cycle wear, not dynamic capacity fade or thermal/calendar aging [1].

B.2 Feedback, chronology and profit

Commitment tracking uses target $g_t^* = Q_t^{\text{sub}}/\Delta_t$ and interval PV. Feasible charge and discharge are bounded by

$$C_t = \min\left(\bar{P}, \frac{[E^{\max} - \rho_t e_t]_+}{\eta_c \Delta_t}\right), \quad D_t = \min\left(\bar{P}, \frac{\eta_d \left[\rho_t e_t - E^{\min}\right]_+}{\Delta_t}\right). \tag{17}$$

It adjusts net battery power toward the grid target, recovers curtailed PV where headroom permits, and enforces connection limits. Interval-average PV feedback is an idealized information assumption. Executed battery state, curtailment and deviations are retained at every interval.

Realized accounting uses

$$\delta_t = g_t^{\text{real}}\Delta_t - Q_t^{\text{sub}}, \tag{18}$$

$$J^{\text{real}} = \sum_{a,t} \pi_{a,t}^{\text{real}} q_{a,t}^{\text{sub}} + \sum_t \left(\lambda_t^+ [\delta_t]_+ - \lambda_t^- [-\delta_t]_+\right) - D^{\text{real}} - I^{\text{real}} - C^{\text{trade,real}}. \tag{19}$$

With a single settlement price this reduces to $\lambda_t \delta_t$. Forecast penalties, tail-risk terms and terminal continuation values are absent from realized cash. Final energy is reported separately. The state and binding positions carry across days; no daily physical reset occurs in a continuous comparison.

C Forecast construction and information timing

C.1 Issue, availability and delivery

For forecast initialization τ_v, lead h and availability A_v, define

$$t_{v,h} = \tau_v + h, \qquad A_v = \tau_v + 6 \text{ hours}, \tag{20}$$

$$v^*(k) = \max\{v : A_v \le k, \text{ required delivery horizon covered}\}. \tag{21}$$

The deterministic production forecast at cutoff k is $V_{0,t}^{(k)} = f_\theta(X_{v^*(k),t}, M)$, where M denotes farm assumptions. For delivery day d, let v_d be the preceding day's 00 UTC ensemble issue. The path input is

$$V_{s,t}^{(k)} = f_\theta(X_{v_d,s,t}, M), \quad s = 1, \dots, 50, \qquad w_s = \frac{1}{51}, \quad s = 0, \dots, 50. \tag{22}$$

The deterministic path updates while the weather-ensemble issue is fixed for the day. Model availability and observation delays are additional eligibility conditions; retrospective fitted models retain their historical-use label.

C.2 Physical and statistical production mappings

For GHI, horizontal beam irradiance BHI and solar zenith θ_z, the physical conversion uses the symbolic relationships

$$G_{\mathrm{DHI}} = G_{\mathrm{GHI}} - G_{\mathrm{BHI}}, \qquad G_{\mathrm{DNI}} \simeq G_{\mathrm{BHI}}/\cos\theta_z, \tag{23}$$

$$G_{\mathrm{POA}} = G_{\mathrm{beam}} + G_{\mathrm{sky}} + G_{\mathrm{ground}}, \tag{24}$$

$$T_{\mathrm{mod}} = T_{\mathrm{air}} + G_{\mathrm{POA}}/(U_0 + U_1 v), \tag{25}$$

$$P_{\mathrm{DC}} = \max\left(0, \frac{P_{\mathrm{AC,rated}}}{\eta_{\mathrm{ref}}} \frac{G_{\mathrm{POA}}}{1000}[1 + \gamma(T_{\mathrm{mod}} - T_{\mathrm{ref}})]\right), \tag{26}$$

$$V_t = \mathbf{1}\{t \text{ outside the seasonal night window}\} \operatorname{clip}\big(\mathcal{C}(P_{\mathrm{DC}}), 0, P_{\mathrm{AC,rated}}\big). \tag{27}$$

Low-sun handling and valid irradiance bounds are required for the DNI conversion. $\mathcal{C}$ represents the selected DC-to-AC/output-loss mapping. The reference coefficients are $U_0 = 25$, $U_1 = 6.84$, $\gamma = -0.004/^\circ\mathrm{C}$ and $\eta_{\mathrm{ref}} = 0.96$; fitted historical conversions retain their own parameters. Tracker position and any rear contribution use explicit geometry assumptions.

A statistical model instead fits normalized target $y_t = P_t/P_{\mathrm{AC,rated}}$:

$$\hat{\theta} = \arg\min_\theta \sum_{d,t} \omega_{dt} \rho\big(f_\theta(X)_t - y_t\big), \qquad \sum_{t \in d} \omega_{dt} = \frac{1}{|\mathcal{D}_{\mathrm{train}}|}. \tag{28}$$

Squared or Huber loss fits point predictions; normalization and preprocessing are learned from training data. Calendar, weather and lead/issue features must be available at the cutoff. A sequence model may use the whole eligible issued weather sequence, including future forecast covariates. Future observed production is reserved for evaluation or an explicit perfect-production reference.

For ensemble values x_s, probabilities w_s and realized power y, marginal quality can be measured by

$$\mathrm{CRPS} = \sum_s w_s |x_s - y| - \frac{1}{2} \sum_{s,r} w_s w_r |x_s - x_r|. \tag{29}$$

RMSE, MAE, bias and coverage provide complementary diagnostics. These scores evaluate forecasts separately from realized profit, physical feasibility and serial dependence. A calibrated collection of marginals requires an explicit temporal coupling before it defines new production paths for storage.

References

[1] B. Xu et al., *Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*, 2017. https://arxiv.org/abs/1707.04567.